

Marmousi2 SLS 100 epoch example

100 轮合成反演结果与局限 | Recorded 9 October 2026

两组实验均已完成 100 轮。固定真实 Q 的 Vp 反演及 Vp/Q 联合反演都降低了数据残差；联合 Q 的恢复仍有明显误差。

Both runs completed 100 epochs. Data residuals decrease, while joint Q recovery remains incomplete.

Fixed true Q / 固定真实 Q

MSE 0.0513675 $\rightarrow$ 0.000436810

Vp RMSE 365.097 $\rightarrow$ 191.772 m/s; Q remains fixed

Joint Vp/Q / 联合反演

MSE 0.0719565 $\rightarrow$ 0.00392416

Vp RMSE 365.097 $\rightarrow$ 206.753 m/s; Q RMSE 99.7373 $\rightarrow$ 82.8195

MSE 为初始和最终固定模型的完整采集评估，已除以固定观测均方能量，未乘训练时的 10^6 。表列 RMSE 覆盖全网格。

MSE is the complete-survey evaluation of one fixed model, normalized by the fixed observation energy, without the training multiplier 10^6 . RMSE above is whole-grid.

单 SLS : Vp 为 6 Hz 相速度，Q 为同频复模量品质因子；密度固定 2000 kg/m³。

Single SLS: Vp is phase speed at 6 Hz; Q is modulus quality factor at 6 Hz. Density is fixed at 2000 kg/m³. This is not a broadband constant-Q model.

117 $\times$ 567 cells, 30 m; 30 shots $\times$ 567 receivers; 15 batches/epoch $\times$ 2 shots; two NVIDIA A30 GPUs.

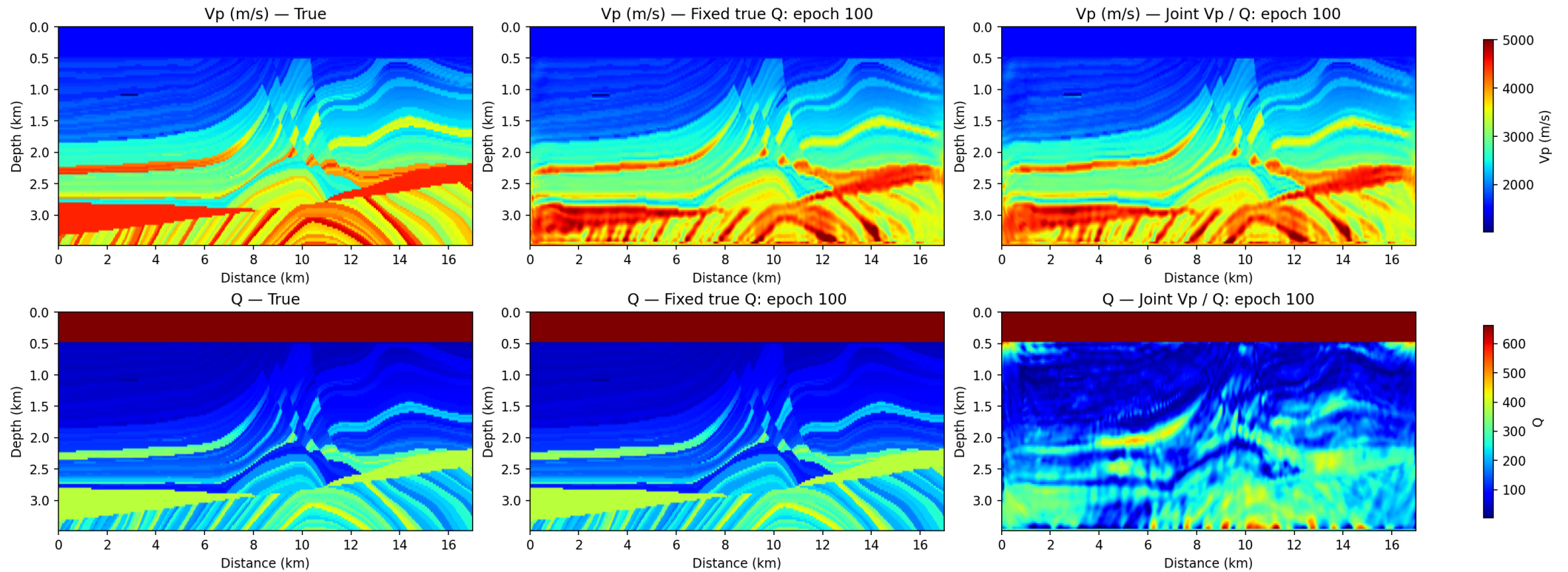
6 Hz Ricker, peak 0.45 s; dt=0.0015 s, nt=4000; order 8, PML=10, full memory, max_vel=6000 m/s.

Adam: Vp lr=10, Q lr=0.5, betas=(0.5,0.99); 1500 updates per run; no learning-rate schedule or explicit model regularization.

True and final models

真实模型与第 100 轮结果 | Shared scales within each parameter row

Marmousi2 SLS — final models with shared scales
Constant density 2000 kg/m³; fixed water Q=1000 exceeds the displayed rock Q scale



Fixed-Q 的 Q 图始终是真值，并非反演结果。水层 Q=1000 超出岩石 Q 色标；联合 Q 可见条带、过估计与欠估计。

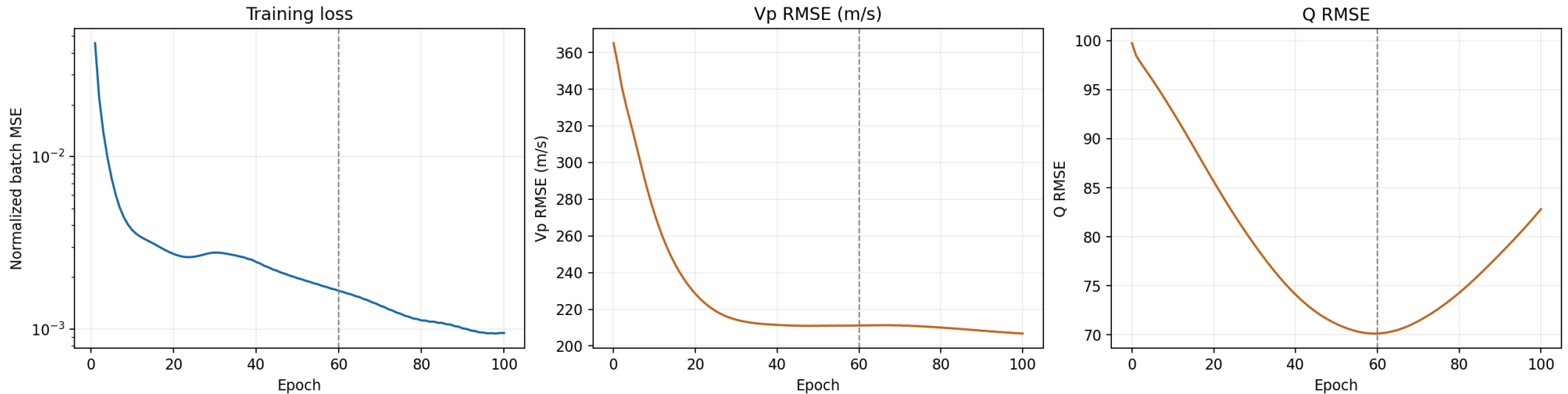
The fixed-Q panel is truth held constant, not recovered Q. Fixed water Q=1000 exceeds the rock-Q color scale. The joint Q field remains distorted.

初始 Vp/Q 采用 Gaussian $\sigma=8$ 网格，恢复并固定前 16 行水层及边框；56,500/66,339 像素可更新。

Gaussian smoothing uses $\sigma=8$ cells; 16 water rows and the outer frame are frozen. 56,500 of 66,339 cells are trainable.

Lower residual does not ensure better Q

数据拟合改善并不保证 Q 恢复改善



Q RMSE 第 59 轮最低 70.131958，第 100 轮为 82.819542；虚线是第 60 轮，最接近最优轮次的已保存模型。

Q RMSE is best at epoch 59 (70.131958), then worsens to 82.819542 at epoch 100. The line marks epoch 60, the nearest stored model checkpoint.

训练曲线平均不同更新前的 batch loss，不能当作一个最终模型的完整采集 loss。初始 Q 不同，也使两组初始 loss 不同。

Training curves average batch losses evaluated before different updates. They are not the complete-survey objective of the final model. The two initial Q models also differ.

真值 RMSE 仅适用于该合成诊断，实测数据不能用未知真实 Q 来择轮次。第 100 轮有 29 个 Q 像素触及下界 5，上界 1200 未触及。

True-model RMSE is available only for this synthetic diagnostic; it is not a field-data stopping criterion. At epoch 100, 29 Q cells reach the lower bound 5; none reach the upper bound 1200.

Interpretation and reproducible evidence

解释边界与可复查证据

Vp-Q 耦合是合理假设，但尚无匹配的 Marmousi 固定 Vp 反 Q 或灵敏度/学习率消融来证明因果。
Vp-Q trade-off is plausible, but no matched Marmousi Q-only or sensitivity/learning-rate ablation isolates it.

有限采集约束、初始平滑、参数尺度、恒定步长和缺少显式正则也可能参与。单凭该结果不能归因为代码错误。
Finite acquisition coverage, initial smoothing, parameter scaling, constant learning rates and absent explicit regularization may also contribute. These results alone do not establish a code defect.

独立 41×41 环绕采集 SLS 小模型：固定真 Vp 反 Q，RMSE $3.099631 \rightarrow 0.114001$ ；联合反演 Q 为 0.993559。其几何和频率不同，不能替代 Marmousi 对照。
The separate 41×41 ring test gives Q-only RMSE $3.099631 \rightarrow 0.114001$, versus joint Q RMSE 0.993559. Its different geometry and frequency prevent transferring this comparison directly.

下载包含初始模型、每 10 轮模型、梯度快照、100 轮 CSV/JSON、独立审阅 Notebook、实验脚本及哈希清单。不含私有算子源码、检查点、机器路径或传输脚本。
The package includes initial/checkpoint arrays, gradient snapshots, 100-epoch CSV/JSON histories, a saved-results notebook, the experiment workflow and hashes. Operator implementations and private runtime material are excluded.

原已执行 Notebook 是 5 轮验证；100 轮来自相同脚本的独立批处理。当前审阅没有重新运行 GPU，亦未添加噪声或检验实测模型失配。
The original executed notebook is a five-epoch validation; 100-epoch evidence comes from separate jobs using the same script. This review did not rerun the GPU. The same-solver synthetic experiment has no added-noise step or field-data mismatch test.

Synthetic Q construction: Dou and Zhang (2016), Chinese Journal of Geophysics 59(11), 4212-4222, section 5.3; doi:10.6038/cjg20161123. The empirical law constructs rock Q truth only; it does not impose a Vp-Q relation during optimization.